

CAROLE MBOMEgni NANA

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Portfolio: <https://carolembomegni.github.io/>

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GitHub: <https://github.com/carolembomegni/>

French – Fluent (Native-level) English – Intermediate (Professional Working Proficiency)

PROFESSIONAL SUMMARY

Data Analyst and Python Developer with over three years of experience delivering data analytics, machine learning, and business intelligence projects. Skilled in Python, SQL, ETL development, data visualization, predictive modeling, and API-based data integration. Experienced in building end-to-end machine learning solutions, including data preparation, feature engineering, model training, evaluation, visualization, and deployment with Streamlit. Familiar with Large Language Models (LLMs), Generative AI, prompt engineering, and AI-assisted workflows. Passionate about leveraging data and AI to solve business problems, automate processes, and support data-driven decision-making.

TECHNICAL SKILLS

Programming & Data Science

- Python: Pandas, NumPy, Scikit-Learn, Matplotlib
- Machine Learning: Classification, Regression, Random Forest, XGBoost, KNN, Logistic Regression
- Deep Learning: TensorFlow, Keras, CNN, Transfer Learning
- Model Training, Evaluation, Feature Engineering and Hyperparameter Optimization

Generative AI & LLMs

- Large Language Models (LLMs)
- Generative AI Applications
- Prompt Engineering
- AI-assisted Research and Analytics
- LLM Output Evaluation
- Grounded and Responsible AI Principles
- Familiarity with LLM API Integration and AI Application Workflows

Data Engineering & APIs

- ETL Pipeline Development
- Python Data Processing
- API Data Extraction and Integration
- JSON Data Processing
- Data Cleaning, Transformation and Validation
- Multi-Source Data Pipelines

Databases & Development Tools

- SQL, SQL Server, MySQL
- Stored Procedures, BULK INSERT, SQL Agent
- Git/GitHub
- Jupyter Notebook, Google Colab, VS Code
- Streamlit

Data Engineering & Cloud

- ETL Pipeline Development
- API Data Extraction and Integration
- Python Data Processing
- JSON Data Processing
- Data Cleaning, Transformation and Validation
- Multi-Source Data Pipelines
- Microsoft Azure Machine Learning (Azure ML Designer)

Business Intelligence & Analytics

- Power BI
- Power Query
- DAX
- Tableau
- Excel
- Data Visualization and Storytelling
- Technical Documentation
- Presentation of Analytical Results to Technical and Non-Technical Audiences

PROFESSIONAL EXPERIENCE

Volunteer – Structured Hospital Environment

Ottawa Civic Hospital – Ottawa, Canada

Since August 2025

- Assisted patients and staff with orientation and daily support services
- Contributed to smooth operational flow in a fast-paced and structured environment

Strengthened organizational, communication, and teamwork skills

Data Analyst

Netview Solutions – Yaoundé, Cameroon

Mai 2021 – April 2024

- Developed Python-based analytical workflows to clean, transform, validate, and analyze structured data from SQL Server, CSV files, and APIs.
- Built and maintained ETL processes integrating data from multiple sources for analytical and reporting purposes.

- Applied statistical analysis and predictive modeling techniques to identify trends, patterns, and business opportunities.
- Developed interactive Power BI and Tableau dashboards to communicate analytical findings and monitor business performance.
- Collaborated with internal stakeholders to translate business requirements into data-driven analytical solutions.
- Produced technical documentation, analytical reports, and presentations for technical and non-technical audiences.
- Performed data quality checks and validation procedures to ensure accurate and reliable analytical outputs.

Junior Software Developer

Resytal – Yaoundé, Cameroon

February 2020 – Decembre 2020

- Developed Python and PHP scripts to automate data processing and application-related workflows.
- Designed, queried, and maintained relational MySQL databases.
- Integrated data flows between internal applications and databases.
- Troubleshoot application and data-processing issues and implemented corrective solutions.
- Optimized database structures and processing workflows to improve reliability and performance

RELEVANT PROJECTS

Brain Tumor MRI Classification & Deployment

- Developed and evaluated deep learning image classification models using Python, TensorFlow, Keras, CNN architectures, and transfer learning.
- Implemented preprocessing, data augmentation, model training, evaluation, and performance analysis workflows.
- Compared model performance using classification metrics and iterative experimentation.
- Deployed the trained model through an interactive
- Streamlit application, creating an end-to-end workflow from model development to user-facing inference.
- Used GitHub for project organization, documentation, and version control.

Application: [Analyse IRM cérébrale · Streamlit](#)

Generative AI Business Research & LLM Evaluation

- Used Large Language Models to conduct structured research, synthesize complex information, and support business analysis.
- Designed and iteratively refined prompts to improve relevance, structure, and quality of AI-generated outputs.
- Evaluated AI-generated responses for accuracy, consistency, relevance, and potential limitations.
- Produced analytical summaries, presentations, and business recommendations for different audiences.

Ontario 511 ETL Data Pipeline (SQL Server)

- Designed and automated a real-time data ingestion and ETL pipeline.
- Developed stored procedures, triggers, logging systems, and scheduled jobs.
- Implemented data quality controls and monitoring processes.
- Created KPI dashboards to support operational performance tracking and reporting.

Dress Sales Prediction (Azure Machine Learning)

- Developed and evaluated predictive machine learning models using Microsoft Azure ML Designer.
- Performed data preprocessing, feature engineering, and model validation.
- Compared classification algorithms and optimized model performance using evaluation metrics.
- Built end-to-end machine learning pipelines for training, testing, and prediction.
- Stored and managed structured datasets using MongoDB.
- Generated visual reports and analytical insights to support business decision-making.

Customer Churn Prediction Analysis

- Cleaned and validated customer datasets using Python and Pandas.
- Performed exploratory data analysis to identify customer behaviour trends.
- Built predictive models using Scikit-Learn.
- Generated business recommendations to improve customer retention.
- Developed dashboards and executive reports to communicate insights.

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Power BI certificate

EDUCATION

- Diploma in Applied Data Science, La Cité College – Ottawa, Ontario
- Bachelor's degree in computer science, University of Yaoundé I, Cameroon
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